Counting the Maximum Blackouts That Keep Every Lattice Rectangle Identifiable

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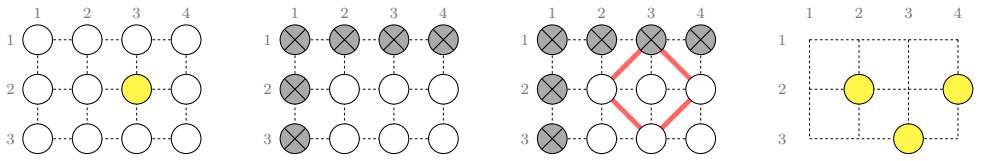
Abstract

Black out points of an $h \times m$ grid so that every lattice rectangle, tilted or not, is still determined by its white corners. Kagey asked for the largest number of black points. We count the blackouts that attain it. On two rows there are $m2^{m-1}$ of them. On three rows we find the exact number, with a closed form, a rational generating function, and the count up to symmetry. On four rows a finite automaton accepts the maximum blackouts. With computer help we obtain their generating function, which is rational with a denominator of degree 420, and the growth rate $3.35199\ldots$ of their number. For every fixed height the generating function of the counts is rational. We also prove a conjecture on the largest entry in each row of the sequence A338885.

1 Introduction

Start with an $h \times m$ grid and black out some of its points. The game master chooses a rectangle (tilted ones are allowed) and tells you which of its corners are white. Can you recover all four corners? Kagey [6, 7] asks how many points can be blacked out while keeping the answer unique for every rectangle. Such a set is a *valid blackout* (Definition 1). One round is shown in Figure 1.

Write $f(h, m)$ for the largest size of a valid blackout, and $a_h(m)$ for the number of valid blackouts of that size. We call them *optima*. The values $f(h, m)$ form the table A397315 of the OEIS [14], and the counts $a_h(m)$ form the table A399580. The companion paper [12] is about f. It proves the bound $f(h, m) \le m + \lfloor h^2/4 \rfloor$ with its cases of equality, shows that equality holds for all large m, finds the order of the least such width, and gives $f(h, m)$ for every $h \le 6$. This paper is about the counts. It uses from the companion paper only the statements listed in Theorem 10, and the companion paper uses nothing from this one.



(a) the point $(3, 2)$ (b) our blackout (c) the rectangle (d) what we are shown

Figure 1: One round on a largest valid blackout of the 3×4 grid. Points are (column, row). Gray crossed circles are black. Panel (c) shows the red rectangle, and (d) its yellow corners. The three yellow corners fix the fourth by Lemma 3.

Put $q_h = \lfloor h^2/4 \rfloor$. We record a blackout by the set of black rows of each column, read from left to right. The results are as follows.

(1) *Two rows.* We have $a_2(m) = m2^{m-1}$ for $m \geq 3$ (A001787). Up to symmetry the optima are counted by A322284, by a bijection with placements of kings (Theorem 11 and Proposition 12).

(2) *Three rows.* We find $a_3(m)$ in closed form (Theorem 21). For $m \geq 4$ it is $4(m-1)(3m+2)\,3^{m-5} + E_m$, where E_m counts the optima with a column black in rows 1 and 3 and has order $m^2(1+\sqrt{3})^m$. So $a_3(m) \sim \frac{4}{81} m^2 3^m$. These sequences are A401223 and A398448. The series $\sum_{m\geq 1} a_3(m) z^m$ is rational with denominator $(1-3z)^3(1-2z-2z^2)^3(1+2z^2)^3$ (Theorem 20). The optima up to symmetry are counted in Theorem 22, which gives A401226.

(3) *Four rows.* For $m \geq 4$ no valid blackout hides all 4 corners of a rectangle (Theorem 23). For $m \geq 6$ a finite automaton with 6 columns of memory accepts exactly the optima (Proposition 26). With computer help we find the generating function of $a_4(m)$ (A401224). It is rational with a reduced denominator of degree 420, and $a_4(m) \sim c\, m^4 \rho_4^m$, where $\rho_4 = 3.35199\ldots$ is algebraic of degree 20 (Theorem 27 and Corollary 28).

(4) *Five rows.* Two transfer counters give $a_5(m)$ for $8 \leq m \leq 23$ (A401225, Table 3).

(5) *Fixed height.* For every h the valid column words form a regular language, $f(h, m) - m$ is ultimately periodic in m, and $\sum_{m\geq 1} a_h(m) z^m$ is rational (Theorems 29 and 30). The proof does not use the companion paper.

(6) *Tilted rectangles.* A tilted lattice rectangle whose bounding box has height n has width at most $\lfloor n^2/4 \rfloor + 1$. This proves a conjecture of Kagey in A338885, that the largest entry of its row n is A033638(n) (Remark 6).

Statements that rest on a computation are marked as computer-assisted, and Section 8 names the programs and the stored evidence.

Related work. The value $f(3, m) = m + 2$ was also proved in Lean by the Omega Institute [16]. The row bound $f(h, m) \leq m + q_h$ of the companion paper is, in graph form, due to Neuwirth [11, Theorem 1(ii) and Proposition 17]. Counting configurations on a grid of fixed height by a transfer matrix is a standard method [15, Section 4.7], [3, Proposition V.6]. Examples are the independent sets of grid graphs [1, 2], the nonattacking kings of Wilf [17], and the dominating sets of grids and king graphs counted by Mertens [9].

2 Definitions and tools

2.1 Blackouts

An $h \times m$ grid has h rows, m columns, and point set V. The point (i, j) lies in column i and row j, with $(1, 1)$ at the top left (Figure 1(a)). A *rectangle* has four grid points as corners and positive area. We write $C(R)$ for the corner set of a rectangle R. It is *axis-aligned* if its sides follow the grid lines, and *tilted* otherwise.

A *blackout* is a set $S \subseteq V$ of black points. The other points are white. The *presentation* of a rectangle R is $P_S(R) = C(R) \setminus S$, the set of its white corners. We say that R *shows* these corners, and that R is *hidden* if it shows none.

Definition 1 (valid blackout)**.** A blackout S is *valid* if distinct rectangles have distinct presentations. We write $f(h, m)$ for the largest size of a valid blackout of the $h \times m$ grid, and $a_h(m)$ for the number of valid blackouts of that size, which we call *optima*. We count them as sets of points, so reflections and rotations are not identified.

Transposing the grid gives $f(h, m) = f(m, h)$ and $a_h(m) = a_m(h)$. A grid with 1 row or 1 column has no rectangles, so $f(1, m) = m$ and $a_1(m) = 1$.

Column i of a blackout has *black rows* $B_i = \{j : (i, j) \in S\}$. It is *empty*, a *singleton*, a *double*, or a *triple* when $|B_i|$ is 0, 1, 2, or 3, and *full* when $|B_i| = h$. The *column word* of S is $B_1 B_2 \cdots B_m$. The *pair set* F of S is the set of pairs of rows that are both black in some column. Throughout, $q_h = \lfloor h^2/4 \rfloor$, which is A002620$(h)$.

Lemma 2 (difference sets)**.** *Two rectangles R_1 and R_2 have the same presentation if and only if $C(R_1) \triangle C(R_2) \subseteq S$.*

Proof. The sets $C(R_1) \setminus S$ and $C(R_2) \setminus S$ agree exactly when every point that lies in just one of $C(R_1)$ and $C(R_2)$ belongs to S. □

Lemma 3 (three corners suffice)**.** *Three corners of a rectangle determine it. Hence two distinct rectangles with the same presentation show at most 2 corners, and a valid blackout hides at most 1 rectangle.*

Proof. Three corners form a right triangle. Its right angle is at the corner q adjacent to the other two, p and r, so the fourth corner is $p + r - q$. Two hidden rectangles would both have the empty presentation. □

2.2 Tilted rectangles

Put $W_h = 1 + \lfloor (h-1)^2/4 \rfloor$, so $W_3 = 2$, $W_4 = 3$, and $W_5 = 5$.

Lemma 4 (leftmost corner)**.** *A tilted rectangle R has a unique corner $\ell = (x, r)$ in its smallest column. The sides from ℓ are $(a, -b)$ and (c, d) with positive integers a, b, c, d and $ac = bd$, and the corners are*

$$(x, r), \qquad (x + a,\, r - b), \qquad (x + c,\, r + d), \qquad (x + a + c,\, r - b + d). \tag{1}$$

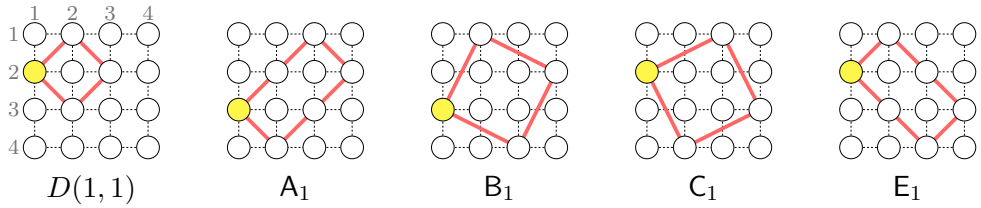


Figure 2: The tilted rectangles of the $4 \times m$ grid, up to translation. Each is drawn in a 4×4 window, and its leftmost corner ℓ is yellow.

The rectangle meets the rows $r - b$ to $r + d$ and the columns x to $x + a + c$. We call $a + c$ its span.

Proof. No side of R is vertical. Hence two corners in the smallest column x would be opposite. The other two corners have the same midpoint, so they would also lie in column x, and R would have no area. The two sides from ℓ therefore have positive first coordinates a and c. They are perpendicular, so their second coordinates have product $-ac$. Thus one side points up and the other down, which gives the form above with $ac = bd$. □

Lemma 5 (span bound)**.** *Every tilted rectangle of the $h \times m$ grid has span $a + c \le W_h$.*

Proof. The rectangle fits in h rows, so $b + d \le h - 1$. Since $(a - 1)(c - 1) \ge 0$, we get $a + c \le ac + 1 = bd + 1$. Finally $bd \le \lfloor (b + d)^2/4 \rfloor \le W_h - 1$. □

Remark 6 (a conjecture in A338885). The same computation settles a conjecture of Kagey. Row n of his triangle A338885 lists all k such that some tilted lattice rectangle touches all 4 sides of an $n \times k$ rectangle, and he conjectures there that the largest such k is A033638$(n) = \lfloor n^2/4 \rfloor + 1$. This is true for every $n \ge 2$. In (1) the bounding box has height $b + d = n$ and width $k = a + c$. As in Lemma 5, $k \le bd + 1 \le \lfloor n^2/4 \rfloor + 1$. For equality take $b = \lfloor n/2 \rfloor$, $d = \lceil n/2 \rceil$, and the perpendicular sides $(1, -b)$ and (bd, d).

The *diamond* $D(i, j)$ is the square with corners $(i, j + 1)$, $(i + 1, j)$, $(i + 2, j + 1)$, and $(i + 1, j + 2)$. The rectangles A_i, B_i, C_i, and E_i have one corner in each of the columns i to $i + 3$, in the rows 3412, 3142, 2413, and 2143 from left to right (Figure 2).

Lemma 7 (tilted rectangles on 3 and 4 rows)**.** *The tilted rectangles of the $3 \times m$ grid are the diamonds $D(i, 1)$. Those of the $4 \times m$ grid are $D(i, 1)$, $D(i, 2)$, A_i, B_i, C_i, and E_i.*

Proof. We use Lemma 4. On 3 rows, $b + d \le 2$ forces $b = d = 1$. Then $ac = 1$, so $a = c = 1$ and ℓ is in row 2. On 4 rows, $b = d = 1$ again gives diamonds, with ℓ in row 2 or 3. Otherwise $\{b, d\} = \{1, 2\}$. Then $ac = 2$, so $\{a, c\} = \{1, 2\}$, and ℓ is in row $1 + b$. The 4 choices of sides, $(2, -2), (1, 1)$, $(1, -2), (2, 1)$, $(2, -1), (1, 2)$, and $(1, -1), (2, 2)$, give A_i, B_i, C_i, and E_i. □

2.3 Axis-aligned rectangles

A blackout S is *axis-valid* if distinct axis-aligned rectangles have distinct presentations. In the *row-column graph* G_S, the vertices are the rows $r_1, \ldots, r_h$ and the columns $c_1, \ldots, c_m$, and r_j is joined to c_i for each black point (i, j).

Lemma 8 (short cycles)**.** *If G_S has no 4-cycle and no 6-cycle, then S is axis-valid. The converse holds for $m \geq 3$.*

Proof. Let $m \geq 3$. A 4-cycle $r_j\, c_i\, r_{j'}\, c_{i'}$ gives the rectangles $\{i, w\} \times \{j, j'\}$ and $\{i', w\} \times \{j, j'\}$ for a third column w. Their corner sets differ in 4 black points. A 6-cycle $r_a\, c_x\, r_b\, c_y\, r_c\, c_z$ gives $\{x, y\} \times \{a, b\}$ and $\{y, z\} \times \{a, c\}$, whose corner sets differ in the 6 black points of the cycle. In both cases Lemma 2 shows that S is not axis-valid.

Conversely, let axis-aligned rectangles $R_1 \neq R_2$ have the same presentation, so that $C(R_1) \triangle C(R_2) \subseteq S$. They do not share opposite corners. Hence they share 0, 1, or 2 corners, and 2 shared corners lie in one row or in one column. We may assume that it is a row a. If they share no corner, then R_1 is hidden, which is a 4-cycle. If they share (x, a) and (y, a), their other rows $b \neq c$ give the 4-cycle $r_b\, c_x\, r_c\, c_y$. If they share just (x, a), write $R_1 = \{x, y\} \times \{a, b\}$ and $R_2 = \{x, z\} \times \{a, c\}$ with $y \neq z$ and $b \neq c$. The other 6 corners are black and form the 6-cycle $r_a\, c_y\, r_b\, c_x\, r_c\, c_z$. □

The next lemma tells when a set of at most 2 points is the presentation of an axis-aligned rectangle.

Lemma 9 (axis-aligned rivals)**.** *Let S hide no axis-aligned rectangle, and let P be a set of at most 2 white points. A* rival *of P is an axis-aligned rectangle with presentation P. The empty set has no rival. Otherwise P has a rival exactly in the following cases.*

- *(a) $P = \{(x, r)\}$, and some $s \in B_x$ has $\{r, s\} \in F$.*
- *(b) $P = \{(x, r), (x, s)\}$ is a vertical pair, and $\{r, s\} \in F$.*
- *(c) $P = \{(x, r), (z, r)\}$ is a horizontal pair, and $B_x \cap B_z \neq \emptyset$.*
- *(d) $P = \{(x, r), (z, s)\}$ is an oblique pair with $x \neq z$ and $r \neq s$, and (x, s) and (z, r) are black.*

Proof. A rival of $P \neq \emptyset$ has rows $\{r, s\}$ and columns $\{x, z\}$, and each of its corners outside P is black. Reading off these corners gives the 4 conditions. In (a) the black corners are (x, s), (z, r), and (z, s), so $s \in B_x$ and column z contains the pair $\{r, s\}$. Here $z \neq x$ holds automatically, since (x, r) is white. □

2.4 What we use from the companion paper

Theorem 10 ([12])**.** *Let $m \geq 3$.*

- *(a)* (Row bound.) *Every axis-valid blackout of the $h \times m$ grid has at most $m + q_h$ points. Hence $f(h, m) \leq m + q_h$.*
- *(b)* (Equality.) *Let $h \geq 4$. A blackout of the $h \times m$ grid with $m + q_h$ points is axis-valid if and only if every column is a singleton or a double, the q_h doubles have distinct row pairs, and these pairs are the edges of a complete bipartite graph on the rows with parts of sizes $\lfloor h/2 \rfloor$ and $\lceil h/2 \rceil$.*
- *(c)* (Values.) *We have $f(2, m) = m + 1$ and $f(3, m) = m + 2$. We have $f(4, 4) = 7$, $f(4, 5) = 8$, and $f(4, m) = m + 4$ for $m \geq 6$. We have $f(5, 5) = 10$, $f(5, 6) = 11$, $f(5, 7) = 12$, and $f(5, m) = m + 6$ for $m \geq 8$.*

In the companion paper the values $f(4,4)$, $f(4,5)$, $f(5,6)$, and $f(5,7)$ rest on computer searches, and $f(5,m) = m+6$ rests on a finite check that is also verified in Lean. We do not need (c) for 2 and 3 rows, since Theorem 11 and Corollary 16 prove these cases again. Table 1 lists $f(h,m)$ and $a_h(m)$ for $h \le 5$.

$f(h,m)$				
$m\backslash h$	2	3	4	5
2	4	4	5	6
3	4	5	6	7
4	5	6	7	8
5	6	7	8	10
6	7	8	10	11
7	8	9	11	12
8	9	10	12	14
9	10	11	13	15
10	11	12	14	16
11	12	13	15	17
12	13	14	16	18
13	14	15	17	19

$a_h(m)$				
$m\backslash h$	2	3	4	5
2	1	12	32	80
3	12	53	224	892
4	32	224	316	1810
5	80	892	1810	8
6	192	3420	26	76
7	448	12704	292	1184
8	1024	45864	2118	4
9	2304	161992	13332	44
10	5120	562912	75398	350
11	11264	1930312	402656	3004
12	24576	6545496	2037848	24706
13	53248	21988784	9894016	193406

Table 1: The values of $f(h,m)$ (A397315) and $a_h(m)$ (A399580) for $h \le 5$. The columns $h = 2$, 3, 4 of the counts are A001787 from $m = 3$, A401223, and A401224, and the column $h = 5$ is A401225 from $m = 8$. The counts $a_4(m)$ and $a_5(m)$ with $m \ge 4$ are computer-assisted.

3 Two rows

On 2 rows every rectangle is axis-aligned, since a tilted rectangle meets at least 3 rows by Lemma 4. A rectangle of the $2 \times m$ grid is therefore given by its 2 columns.

Theorem 11 (Strip Theorem)**.** *Let $m \ge 3$.*

(a) A blackout of the $2 \times m$ grid is valid if and only if at most 1 of its columns is full.

(b) We have $f(2,m) = m+1$, and the optima are the blackouts with 1 full column and 1 black point in every other column.

(c) We have $a_2(m) = m\,2^{m-1}$. So row 2 of A399580 is A001787 from $m = 3$ on.

For $m = 2$ there is 1 rectangle, so $f(2,2) = 4$ and $a_2(2) = 1$.

Proof. (a) Take two distinct rectangles, with column sets $\{p,q\}$ and $\{p',q'\}$. Their corner sets differ exactly in the points of the columns that belong to one of these sets and not to the other. There are 2 or 4 such columns. Conversely, for any two columns $p \ne q$ choose a third column w. Here we use $m \ge 3$. The rectangles on $\{w,p\}$ and $\{w,q\}$ differ exactly in the points of columns p and q. By Lemma 2, a blackout is valid if and only if no 2 columns are full.

(b, c) A valid blackout keeps a white point in all but at most 1 column, so $|S| \le 2m - (m-1) = m+1$. Equality requires 1 full column and exactly 1 black point in each

other column. There are m choices for the full column and 2 choices in each of the other $m-1$ columns. □

The $2\times m$ grid has 4 symmetries, generated by the row reflection and the column reversal. We count the optima up to these.

Proposition 12 (orbits and kings)**.** *For $m \geq 3$ the number of optima of the $2 \times m$ grid up to symmetry is*

$$m\,2^{m-3} + [m \text{ odd}]\,2^{(m-3)/2}.$$

It equals A322284*$(m-1)$, the number of ways to place $m-1$ nonattacking kings on a $2 \times (2m-2)$ board up to symmetry.*

Proof. We use Burnside's lemma. The identity fixes all $m2^{m-1}$ optima, and the row reflection fixes none, since it moves every singleton. The column reversal and the half turn can fix an optimum only when m is odd and the full column is the middle one. Each of them then fixes $2^{(m-1)/2}$ optima, since the singleton columns come in mirror pairs and one column of each pair determines the other. Dividing the sum by 4 gives the formula.

For the kings, put $n = m-1$. Two kings on a board with 2 rows attack each other exactly when their columns are equal or adjacent. So a placement of n nonattacking kings on the $2 \times 2n$ board consists of columns $c_1 < \cdots < c_n$ with $c_{j+1} \geq c_j + 2$, and a row r_j for each king. The numbers $c_1 - 1$, $c_{j+1} - c_j - 2$ for $1 \leq j < n$, and $2n - c_n$ are nonnegative and sum to 1. Hence exactly one of these $n+1$ numbers is 1. If it is the eth, then $c_j = 2j - 1 + [j \geq e]$. Send this placement to the optimum with full column e and singleton rows $r_1, \ldots, r_n$ from left to right. This is a bijection. It commutes with the row reflection. The column reversal of the board sends c to $2n+1-c$ and reverses the order of the rows r_j. Since $2n+1-c_{n+1-j} = 2j-1+[j \geq n+2-e]$, it sends e to $n+2-e = m+1-e$, as the column reversal of the grid does. Hence the orbits correspond. □

4 Three rows

Take $m \geq 3$ and number the rows $1, 2, 3$ downward. We write a column as a letter. The singletons are $a = \{1\}$, $b = \{2\}$, and $c = \{3\}$, the doubles are $A = \{1, 2\}$, $B = \{1, 3\}$, and $C = \{2, 3\}$, and the triple is $X = \{1, 2, 3\}$. We call a and c the *outer* letters. A blackout without empty columns is then a word in these 7 letters. We also write A, B, C for the corresponding pairs of rows, so that the pair set F is a set of letters.

4.1 The candidates and the window rules

Proposition 13 (the two types)**.** *Let $m \geq 3$.*

(a) A valid blackout of the $3 \times m$ grid has at most $m+2$ points. If it has $m+2$ points, then its word consists of singletons and either one X (type I) *or two different doubles* (type II)*.*

(b) *For every word of type I or II, the graph G_S is a tree. Hence the blackout is axis-valid and hides no axis-aligned rectangle.*

(c) *There are $m3^{m-1} + \binom{m}{2} \cdot 6 \cdot 3^{m-2} = m^2 3^{m-1}$ words of type I or II.*

Proof. (a) Let S be valid. By Lemma 8, G_S has no 4-cycle and no 6-cycle. So no 2 columns share 2 black rows. Suppose first that some column is a triple. Then every other column has at most 1 black point, and $|S| \le 3 + (m-1)$, with equality only for type I. Suppose next that no column is a triple. The doubles have different row pairs. Three doubles would carry all 3 pairs, and their columns would form a 6-cycle with the 3 rows. So there are at most 2 doubles and $|S| \le 4 + (m-2)$, with equality only for type II.

(b) The nonsingleton columns and the 3 rows form a path or a star, and each singleton column is a leaf. Apply Lemma 8. A hidden axis-aligned rectangle would be a 4-cycle.

(c) Place the triple, or an ordered pair of different doubles in 2 of the columns, and choose each singleton freely. □

We call the words of type I or II *candidates*. The only tilted rectangles are the diamonds $D(i,1)$ (Lemma 7). We say that the diamond $D(i,1)$ *fails* in a candidate if it shows 1 or 2 corners and this presentation has a rival in the sense of Lemma 9. Otherwise it *passes*.

Lemma 14 (diamond test)**.** *A candidate is valid if and only if every diamond $D(i,1)$ passes.*

Proof. Axis-aligned rectangles have distinct presentations by Proposition 13(b). Two diamonds $D(i,1)$ and $D(i',1)$ with $i < i'$ share at most 1 corner, which is a point of row 2. If they had the same presentation, the points of rows 1 and 3 in the columns $i+1$ and $i'+1$ would be black by Lemma 2. Then both columns would be B or X, and a candidate has at most 1 such column. So the candidate is valid if and only if no diamond has the presentation of an axis-aligned rectangle. A diamond that shows at least 3 corners is safe by Lemma 3. A hidden diamond is safe because no axis-aligned rectangle is hidden. The remaining diamonds are the subject of Lemma 9. □

A *window* is a factor of 3 consecutive letters $\ell\, t\, r$ of a candidate, in the columns i, $i+1$, and $i+2$. It *passes* or *fails* with $D(i,1)$. The corners of $D(i,1)$ are the *tips* $(i,2)$ and $(i+2,2)$ and the points $(i+1,1)$ and $(i+1,3)$ of the middle column. A tip is black exactly when its letter is b, A, C, or X.

Lemma 15 (window rules)**.** *Let $\ell\, t\, r$ be a window of a candidate with pair set F.*

(a) If t is a singleton, the window fails if and only if both tips are black and $B \in F$.

(b) If $t = A$, the window fails if and only if both tips are black, or exactly 1 tip is black and the letter of the white tip is c or B. If $t = C$, the same holds with a in place of c.

(c) If t is B or X, the window fails if and only if both tips are white with the same letter, or exactly 1 tip is black and the letter of the white tip is a with $A \in F$ or c with $C \in F$.

Proof. (a) At most 1 point of the middle column is black. If both tips are white, at least 3 corners show. Suppose that exactly 1 tip is black and the other tip $(x,2)$ is white. For $t = b$ again 3 corners show. For $t = a$ the diamond shows the oblique pair $(x,2)$, $(i+1,3)$. A rival

would need $(i+1,2)$ to be black (Lemma 9(d)), and it is white. The case $t = c$ is the same. Suppose finally that both tips are black. For $t = b$ the diamond shows the vertical pair in rows 1 and 3, which has a rival if and only if $B \in F$ (Lemma 9(b)). For $t = a$ it shows only $(i+1,3)$, and $B_{i+1} = \{1\}$. By Lemma 9(a) there is a rival if and only if $B \in F$. The case $t = c$ is the same.

(b) Let $t = A$. Then $(i+1,1)$ and $(i+1,2)$ are black and $(i+1,3)$ is white. If both tips are white, 3 corners show. If both are black, the diamond shows only $(i+1,3)$. The candidate has type II, so F contains a second pair, and this pair contains row 3. Its other row is black in column $i+1$, so there is a rival by Lemma 9(a). If exactly 1 tip is black and $(x,2)$ is the white tip, the diamond shows the oblique pair $(x,2)$, $(i+1,3)$. Since $(i+1,2)$ is black, there is a rival if and only if $(x,3)$ is black, which means that the letter of column x is c or B. The case $t = C$ follows by reflecting the rows.

(c) Now both points of the middle column are black. If both tips are black, the diamond is hidden and passes. A white tip has the letter a or c, because the letters A, C have black tips and the word has only one letter B or X. If both tips are white, the diamond shows a horizontal pair, and by Lemma 9(c) there is a rival if and only if the 2 letters are equal. If exactly 1 tip is black, the diamond shows only the white tip $(x,2)$, and $B_x = \{s\}$ with $s = 1$ or $s = 3$. By Lemma 9(a) there is a rival if and only if $\{2,s\} \in F$. □

Corollary 16. *For $m \geq 3$ the word Xa^{m-1} is valid, so $f(3,m) = m+2$, and the optima of the $3 \times m$ grid are the candidates in which every window passes.*

Proof. In Xa^{m-1} every window has the middle letter a and a white right tip. It passes by Lemma 15(a). Now use Proposition 13(a) and Lemma 14. □

We sort the candidates into 7 *classes* by their nonsingleton letters read from left to right, namely X, AC, CA, AB, BA, BC, and CB. We write AC_m for the set of valid words of width m in the class AC, and likewise for the other classes. The row reflection exchanges a with c and A with C, and it fixes b, B, and X. The column reversal reverses the word. Both preserve validity. Hence

$$|AC_m| = |CA_m| \qquad \text{and} \qquad |AB_m| = |BA_m| = |CB_m| = |BC_m|. \tag{2}$$

Table 2 lists the failing windows in the classes AC, AB, and X. It is read off from Lemma 15.

4.2 The class AC

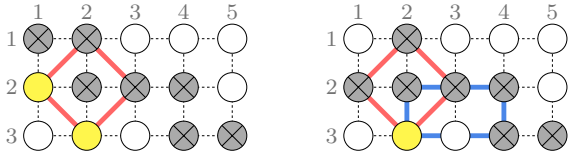
Set $G_{AC}(z) = \sum_{m \geq 2} |AC_m| z^m$, and define G_{AB} and G_X in the same way, where the sum for G_X starts at $m = 1$. Write $\Phi = 1 - 3z$. The nonempty singleton strings with a given first letter have the series z/Φ, and so do those with a given last letter.

Proposition 17 (the class AC)**.** *We have*

$$G_{AC}(z) = \frac{z^2 - 4z^3 + 10z^4 - 8z^5 - 3z^6}{(1-3z)^3}, \tag{3}$$

class	middle	failing windows
AC	A	bAb, bAc, cAb, bAC, cAC
	C	bCb, bCa, aCb, ACb, ACa
AB	singleton s	bsb, bsA, Asb
	A	bAb, bAc, cAb, bAB
	B	aBa, cBc, aBb, bBa, ABa
X	singleton s	bsb, bsX, Xsb
	X	aXa, cXc, aXb, bXa, bXc, cXb

Table 2: The failing windows, by class and middle letter. Here s is any singleton. In the class AC every window with a singleton in the middle passes.



(a) $aAbCc$ is valid (b) $bAbCc$ fails

Figure 3: The diamond test. (a) In $aAbCc$ the diamond $D(1,1)$ shows $(1,2)$ and $(2,3)$. This pair has no rival, since $(1,3)$ is white. (b) In $bAbCc$ it shows only $(2,3)$, and so does the blue rectangle. The window bAb fails.

and

$$|AC_m| = 2(m-1)(3m+2)\,3^{m-5} \qquad (m \ge 4). \tag{4}$$

Proof. Write a word of the class as $UAVCW$ with singleton strings U, V, and W. By Table 2 only the windows centered on A and on C matter. Read C as b when it is a tip, and likewise A. Then the window sAt fails exactly for $st \in \{bb, bc, cb\}$, and sCt fails exactly for $st \in \{bb, ba, ab\}$. We count by the length of V.

If V is empty, a nonempty U must end in a and a nonempty W must start with c. This case contributes $z^2(1+z/\Phi)^2$.

If V is a single letter t, let p_t be the number of letters allowed at the end of U and q_t the number allowed at the start of W. Then $(p_a, p_b, p_c) = (3,1,2)$ and $(q_a, q_b, q_c) = (2,1,3)$. The contribution is

$$z^3 \sum_t \Big(1+\frac{p_t z}{\Phi}\Big)\Big(1+\frac{q_t z}{\Phi}\Big) = z^3\Big(3+\frac{12z}{\Phi}+\frac{13z^2}{\Phi^2}\Big).$$

If V has at least 2 letters, its first letter and the last letter of U form one of 6 allowed pairs, or U is empty and the first letter of V is free. This gives the factor $3+6z/\Phi$. The other end gives the same factor, and the letters of V between its first and last are free. The contribution is $z^4(3+6z/\Phi)^2/\Phi = 9z^4(1-z)^2/\Phi^3$.

Adding the 3 cases gives (3). Since $[z^m]z^k\Phi^{-3} = \binom{m-k+2}{2}3^{m-k}$, the coefficient of z^m for $m \ge 4$ is (4). □

Let E_m be the number of optima of the $3 \times m$ grid with a column that is black in rows 1 and 3 (A398448). Such a column is B or X. So E_m counts the optima in the 5 classes other than AC and CA, and by (2),

$$a_3(m) = 2|AC_m| + 4|AB_m| + |X_m| = 2|AC_m| + E_m \qquad (m \ge 3). \tag{5}$$

4.3 The classes AB and X

Put $L = 1-2z-2z^2$, $N = 1+2z^2$, and $\Psi = LN = 1-2z-4z^3-4z^4$. Call a singleton string *restricted* if it has no 2 letters b at distance 2. In the classes AB and X the windows

bsb fail, so every run of singletons is restricted. Give the letter b the type 1 and the outer letters the type 0. For types i and j, let $h_{ij}(z)$ count by length the strings $y_1 \cdots y_n$ with $n \geq 0$ that can follow 2 letters of types i and j in a restricted string. Write $h_{ij} = h_{ij}^0 + h_{ij}^1$, where h_{ij}^β counts the strings whose last letter has type β, and the empty string is given the type j. The letter y_1 can be a or c, and it can be b when $i = 0$. Hence, for $\beta \in \{0, 1\}$,

$$h_{ij}^\beta = [j = \beta] + 2z\, h_{j0}^\beta + [i = 0]\, z\, h_{j1}^\beta.$$

Solving these 2 linear systems gives

$$\begin{array}{llll} \Psi h_{00}^0 = 1 + 2z^2 + 2z^3, & \Psi h_{00}^1 = z + z^2, & \Psi h_{10}^0 = 1, & \Psi h_{10}^1 = 2z^2 + 2z^3, \\ \Psi h_{00} = 1 + z + 3z^2 + 2z^3, & \Psi h_{01} = 1 + z, & \Psi h_{10} = 1 + 2z^2 + 2z^3, & \Psi h_{11} = 1. \end{array} \tag{6}$$

We read a run of singletons away from the double or triple next to it, as $w_1 w_2 \cdots w_n$ with $n \geq 1$. We need 4 series for such runs. Let g_0 count the restricted runs whose first letter is a given outer letter, and g_1 those with $w_1 = b$. Let u and v count the same runs under the extra condition that w_2, if present, is outer. A missing letter before w_1 acts as a letter of type 0, and the extra condition acts as a letter of type 1. Hence

$$g_0 = z h_{00}, \qquad g_1 = z h_{01}, \qquad u = z h_{10}, \qquad v = z h_{11}. \tag{7}$$

Proposition 18 (the class X)**.** *We have*

$$G_X(z) = z\big(1 + 4u + 2v + 2u^2 + v^2\big) = \frac{z(1 + 2z - 7z^2 - 4z^3 - 2z^4 - 8z^5 - 4z^6)}{L^2 N}.$$

Proof. Write the word as UXW. Both runs are restricted. The failing windows bsX and Xsb of Table 2 say that the second letter from X is outer. The window centered on X allows the neighbor pairs ac, ca, and bb. So a run on one side only contributes $2u+v$, for either side, and runs on both sides contribute $2u^2 + v^2$. The second form follows from (6) and (7). □

Now write a word of the class AB as $UAVBW$. For a letter t next to A in V, let ℓ_t count the strings U that are allowed, including the empty one. For a letter t next to B in V, let r_t count the strings W that are allowed. By Table 2, the window bsA makes the second letter of U from A outer, and nothing restricts the second letter of W. The windows centered on A allow the letters a, b, c before At for $t = a$, only a for $t = b$, and a, c for $t = c$. The windows centered on B allow only c after tB for $t = a$, the letters b, c for $t = b$, and a, b for $t = c$. Hence

$$(\ell_a, \ell_b, \ell_c) = (1+2u+v,\ 1+u,\ 1+2u), \qquad (r_a, r_b, r_c) = (1+g_0,\ 1+g_0+g_1,\ 1+g_0+g_1). \tag{8}$$

Proposition 19 (the class AB, P_B by computer algebra)**.** *With the series of* (6) *to* (8)*,*

$$\begin{aligned} G_{AB}(z) = {} & z^2(1 + 2u)(1 + g_0 + g_1) + z^3(\ell_a r_a + \ell_b r_b + \ell_c r_c) \\ & + z^4 \sum_{\sigma \in \{0,1\}} \lambda_\sigma \big(h_{\sigma 0}^0\,(r_a + r_c) + 2h_{\sigma 0}^1\, r_b\big), \end{aligned} \tag{9}$$

where $\lambda_0 = \ell_a + \ell_c$ *and* $\lambda_1 = \ell_b$*. Moreover* $G_{AB} = P_B/\Psi^3$ *with*

$$P_B(z) = z^2(1 + z)(1 + z^2 - 3z^3 + 4z^6 + 4z^7). \tag{10}$$

Proof. Let k be the length of $V = v_1 \cdots v_k$. For $k = 0$ the windows sAB and ABt of Table 2 require $s \neq b$ and $t \neq a$, which gives the first term. For $k = 1$ the window Av_1B passes, and the windows centered on A and B give $\ell_t r_t$ for $v_1 = t$.

Let $k \geq 2$. The window Av_1v_2 makes v_2 outer, and the window $v_{k-1}v_kB$ passes. So V is a restricted string whose second letter is outer, and its weight is $\ell_{v_1} r_{v_k}$. Let σ be the type of v_1. Summing ℓ_{v_1} over the letters v_1 of type σ gives λ_σ. The string $v_3 \cdots v_k$ follows 2 letters of types σ and 0. If it is empty, then $v_k = v_2$ is outer and the 2 choices give $r_a + r_c$. If it is not empty and v_k is outer, there are 2 choices for v_2, and $h^0_{\sigma 0}$ counts each string $v_3 \cdots v_{k-1}$ twice, once for each v_k. So the weight is again $h^0_{\sigma 0}(r_a + r_c)$. If $v_k = b$, the 2 choices for v_2 give $2h^1_{\sigma 0} r_b$. This proves (9). Substituting (6) and (7) and expanding gives (10). □

4.4 Generating function, closed form, and asymptotics

Theorem 20 (three-row generating function, P by computer algebra)**.**

(a) We have

$$\sum_{m \geq 1} a_3(m) z^m = 2G_{AC} + 4G_{AB} + G_X = \frac{P(z)}{Q(z)}, \tag{11}$$

where $Q = (1-3z)^3(1-2z-2z^2)^3(1+2z^2)^3$ *and*

$$\begin{aligned} P(z) = z &- 3z^2 - 34z^3 + 228z^4 - 641z^5 + 1307z^6 - 2186z^7 \\ &+ 2056z^8 - 236z^9 - 1060z^{10} + 2584z^{11} - 4264z^{12} \\ &- 1440z^{13} + 704z^{14} - 1024z^{15} + 2080z^{16} + 2176z^{17} + 384z^{18}. \end{aligned}$$

(b) The polynomials P and Q are coprime. So Q is the reduced denominator, and $a_3(m)$ satisfies the linear recurrence of order 15 *with characteristic polynomial $x^{15}Q(1/x)$ for $m \geq 19$ and no recurrence of smaller order.*

(c) We have

$$\sum_{m \geq 3} E_m z^m = \frac{z^3 \, P_E(z)}{(1-2z-2z^2)^3(1+2z^2)^3},$$

where $P_E = 43 - 90z + 128z^2 - 344z^3 + 44z^4 + 80z^5 + 328z^6 + 1728z^7 + 1888z^8 + 2208z^9 + 2016z^{10} + 640z^{11}$.

Proof. (a) For $m \geq 3$ the coefficient of z^m in $2G_{AC} + 4G_{AB} + G_X$ is $a_3(m)$ by (5). The coefficients of z and z^2 are 1 and 12. These are $a_3(1)$ and $a_3(2) = a_2(3)$, by Section 2 and Theorem 11. Bringing the 3 series of Propositions 17, 18, and 19 to the common denominator Q gives P.

(b) The factors Φ, L, and N of Q are irreducible and distinct. At $z = \frac{1}{3}$ only G_{AC} has a pole, and the numerator of (3) takes the value $\frac{4}{81}$ there. So the pole at $\frac{1}{3}$ has order 3. The remainders of P_B modulo L and modulo N are $\frac{z}{2}$ and $\frac{1}{4} - \frac{z}{2}$. They are not zero, so G_{AB} has poles of order 3 at the roots of L and of N. The series G_X has poles of order at most 2 there, and G_{AC} has none. Hence the series (11) has a pole of order 3 at every root of Q, and P and Q are coprime. The statement on recurrences follows, since $\deg P = 18$.

(c) By (5) the series is $4G_{AB} + G_X$ without its terms in z and z^2, which are $z + 10z^2$. □

The polynomials P, P_B, and P_E were expanded in exact arithmetic with SymPy [10]. Let $\rho = 1 + \sqrt{3}$ and $\sigma = 1 - \sqrt{3}$, so that $L = (1 - \rho z)(1 - \sigma z)$.

Theorem 21 (the number of three-row optima, checked by computer algebra)**.** *For every $m \geq 4$,*

$$a_3(m) = 4(m-1)(3m+2)\,3^{m-5} + p_+(m)\,\rho^m + p_-(m)\,\sigma^m + (-2)^{\lfloor m/2 \rfloor}\kappa(m), \tag{12}$$

where

$$432\,p_\pm(m) = (19 \pm 11\sqrt{3})m^2 + (61 \pm 41\sqrt{3})m + 104 \pm 28\sqrt{3},$$

and $\kappa(m) = (2m^2 + 41m - 104)/216$ for even m and $\kappa(m) = (7m^2 - 32m - 175)/432$ for odd m. The last 3 terms are E_m, and

$$E_m = p_+(m)\,\rho^m + p_-(m)\,\sigma^m + (-2)^{\lfloor m/2 \rfloor}\kappa(m) \tag{13}$$

holds for every $m \geq 2$. Moreover $E_m > 0$,

$$E_m \sim \frac{19 + 11\sqrt{3}}{432}\,m^2(1+\sqrt{3})^m, \qquad a_3(m) = \frac{4}{81}m^2 3^m - \frac{4}{243}(m+2)3^m + O\big(m^2(1+\sqrt{3})^m\big), \tag{14}$$

and the fraction of the $m^2 3^{m-1}$ candidates that are valid tends to $\frac{4}{27}$.

Proof. By (5) and (4) we have $a_3(m) = 4(m-1)(3m+2)3^{m-5} + E_m$ for $m \geq 4$. So it is enough to prove (13). By (5), E_m is the coefficient of z^m in $4G_{AB} + G_X$ for $m \geq 3$. This also holds for $m = 2$. Indeed the optima of the 3×2 grid are the transposes of the optima of Theorem 11. They are the 12 words of type I or II, and 10 of them contain B or X. By Propositions 18 and 19, the series $4G_{AB} + G_X$ has the denominator L^3N^3 of degree 12 and a numerator of degree 13. So E_m satisfies for $m \geq 14$ the linear recurrence of order 12 with characteristic polynomial $x^{12}L(1/x)^3N(1/x)^3$. Its roots are ρ, σ, and $\pm i\sqrt{2}$, each of multiplicity 3. The right side of (13) is a combination of $m^j\rho^m$, $m^j\sigma^m$, and $m^j(\pm i\sqrt{2})^m$ with $j \leq 2$, so it satisfies the same recurrence for all m. The two sides agree for $2 \leq m \leq 13$, which we checked in exact arithmetic. Hence they agree for all $m \geq 2$.

We have $|\sigma| < 1$ and $|(-2)^{\lfloor m/2 \rfloor}\kappa(m)| = O(m^2 2^{m/2})$, and $\sqrt{2} < \rho$. This gives the asymptotic form of E_m. The expansion of $a_3(m)$ follows, since $4(m-1)(3m+2)3^{m-5} = \frac{4}{81}m^2 3^m - \frac{4}{243}(m+2)3^m$. Finally $E_m > 0$ because Xa^{m-1} is an optimum by Corollary 16, and $E_2 = 10$. □

4.5 Optima up to symmetry

Let $o_3(m)$ be the number of optima of the $3 \times m$ grid up to the row reflection and the column reversal. Its first values for $m = 3, \ldots, 8$ are 15, 58, 227, 864, 3190, and 11502 (A401226).

Theorem 22 (three-row orbits)**.** *Let* $M = 1 - 2z^2 - 2z^4$*. Then*

$$4\sum_{m\geq 3} o_3(m)z^m = \sum_{m\geq 3} a_3(m)z^m + z^3 + \frac{4z^3(1+z^4+z^6)}{M\,(1+2z^4)} + \frac{2z^3(1-z)(1+5z+3z^2-3z^3)}{(1-3z^2)^2}.$$

For even $m \geq 4$ *we have* $o_3(m) = \frac{1}{4}\big(a_3(m) + 2m\,3^{m/2-2}\big)$.

Proof. By Burnside's lemma, $4o_3(m)$ is $a_3(m)$ plus the numbers of optima fixed by the row reflection, the column reversal, and the half turn.

Row reflection. A fixed word uses only b, B, and X. A candidate of this kind is b^iXb^j. For $i \geq 2$ the window bbX fails, and for $j \geq 2$ the window Xbb fails (Table 2). So the only fixed optimum is bXb, which gives the term z^3.

Column reversal. A fixed word is a palindrome. A candidate has different doubles, so a fixed candidate has the form UXU^{rev} and m is odd. The window centered on X requires the last letter of U to be b. As in Proposition 18, the allowed strings U are counted by v. This gives $z\,v(z^2)$.

Half turn. It reverses the word and exchanges a with c and A with C. A fixed word of type I is UXU', where U' is the image of U. The neighbors of X are then ac, ca, or bb, which are the allowed pairs. So these words give $z\,(2u+v)(z^2)$. A fixed word of type II has A and C in mirror columns j and $m+1-j$. Put $m = 2k$ or $m = 2k+1$, where in the second case the middle letter is b. Take the class AC, so $j \leq k$. The first k letters determine the word. Only the windows centered on A and C can fail, and these 2 windows are mirror images. If $j = 1$, the window on A does not exist or does not restrict the first k letters, and there are 3^{k-1} words. If $j = k \geq 2$, the right neighbor of A is C or the middle b, so the left neighbor is a and there are 3^{k-2} words. If $1 < j < k$, there are 6 allowed pairs of neighbors and $2 \cdot 3^{k-2}$ words. The sum over j is $2k\,3^{k-2}$ for $k \geq 2$. The class CA has the same count. So type II gives 2 fixed optima for $m = 3$ and $4k\,3^{k-2}$ for $m \geq 4$, with series

$$2z^3 + \tfrac{4}{9}(1+z)\Big(\frac{3z^2}{(1-3z^2)^2} - 3z^2\Big) = \frac{2z^3(1-z)(1+5z+3z^2-3z^3)}{(1-3z^2)^2}.$$

By (6) and (7) the 2 contributions of the words UXU^{rev} and UXU' add up to $2z\,(u+v)(z^2) = 4z^3(1+z^4+z^6)/\Psi(z^2)$, and $\Psi(z^2) = M(1+2z^4)$. For even $m \geq 4$ only the type II words are fixed, and $4k\,3^{k-2} = 2m\,3^{m/2-2}$. □

5 Four rows

Let $m \geq 6$. Then $f(4,m) = m+4 = m+q_4$ by Theorem 10(c). So every optimum is an axis-valid blackout with $m+q_4$ points, and Theorem 10(b) describes it. It has 4 doubles and $m-4$ singletons, and the row pairs of the doubles are the 4 edges of a 4-cycle on the rows. There are 3 such cycles, one for each of the bipartitions $\{1,2\} \mid \{3,4\}$, $\{1,3\} \mid \{2,4\}$,

and $\{1,4\} \mid \{2,3\}$. A *candidate* is a word of singletons and doubles in which the doubles carry each edge of one of these cycles exactly once. Its pair set F is the edge set of the cycle. There are $3\,(m)_4\,4^{m-4}$ candidates, where $(m)_4 = m(m-1)(m-2)(m-3)$. Every candidate is axis-valid, and it hides no axis-aligned rectangle, since its doubles have different pairs. By Lemma 7, each tilted rectangle has span at most 3.

5.1 Hidden rectangles

Theorem 23 (no hidden rectangle)**.** *For $m \geq 4$, no valid blackout of the $4 \times m$ grid hides a rectangle.*

Proof. Suppose that a rectangle R is hidden. We find two other rectangles $R_1 \neq R_2$ with $C(R) = C(R_1) \triangle C(R_2)$. By Lemma 2 they have the same presentation, so the blackout is not valid. If R is axis-aligned with columns x, y, take a third column w and let R_1 and R_2 have the rows of R and the columns $\{x, w\}$ and $\{y, w\}$. For the tilted rectangles, let $1 \leq i \leq m-3$ and let Q_i be the axis-aligned rectangle with corners $(i,2)$, $(i,3)$, $(i+3,2)$, and $(i+3,3)$. Writing a rectangle for its corner set, we have

$$\begin{aligned} \mathsf{C}_i = \mathsf{A}_i \triangle \mathsf{Q}_i, \qquad & \mathsf{E}_i = \mathsf{B}_i \triangle \mathsf{Q}_i, \\ D(i+1,1) = D(i,2) \triangle \mathsf{A}_i, \qquad & D(i+1,2) = D(i,1) \triangle \mathsf{E}_i. \end{aligned} \tag{15}$$

Each of the 3 sets in an identity is the symmetric difference of the other 2. The first line handles A_i, B_i, C_i, and E_i. For a diamond $D(j,1)$ or $D(j,2)$, use the second line with $i = j-1$ if $j \geq 2$, and with $i = j$ if $j = 1$. In the second case we need the column $j+3 = 4$, and $m \geq 4$. □

The condition $m \geq 4$ is needed. On the 4×3 grid the blackout $\{(1,2),(2,1),(3,2),(2,3)\}$ is valid and hides $D(1,1)$.

Remark 24 (forbidden-block bound). A count of excluded blocks also bounds $a_4(m)$. Let $m \geq 6$, and call 4 consecutive singleton columns of an optimum a *block*, written as the word of its black rows. For each pair set F at least 44 of the 256 words cannot occur. They are the words 2413, 3412, 2143, and 3142 of the corner rows of C_i, A_i, E_i, and B_i, the 16 other words obtained from these by exchanging 2 letters, and at least 24 words obtained from them by changing 1 letter r to s with $\{r,s\} \in F$. Since at least $\lfloor m/4 \rfloor - 4$ disjoint blocks exist, this gives

$$a_4(m) \leq 3\,(m)_4\,4^{m-4}\Big(\frac{53}{64}\Big)^k, \qquad k = \max\big(0, \lfloor m/4 \rfloor - 4\big),$$

so $a_4(m) = O\big(m^4\,212^{m/4}\big)$ and $212^{1/4} < 3.816$. We omit the proof, since Corollary 28 gives the exact rate. The script `four_row_block_words.py` lists the excluded words and checks them on all optima with $m \leq 10$.

5.2 The automaton

Lemma 25 (six columns of memory)**.** *Let $m \geq 6$. A candidate of the $4 \times m$ grid is an optimum if and only if*

(i) no tilted rectangle is hidden,

(ii) no tilted rectangle that shows 1 or 2 corners has a rival in the sense of Lemma 9, and

(iii) no 2 distinct tilted rectangles that lie in 7 consecutive columns have the same presentation.

Proof. An optimum is valid, so (ii) and (iii) hold, and (i) holds by Theorem 23. Conversely, let a candidate satisfy (i) to (iii), and suppose that 2 distinct rectangles have the same presentation P. They are not both axis-aligned, since a candidate is axis-valid. If P is empty, both are hidden. One of them is tilted, which contradicts (i). So P is not empty, and $|P| \leq 2$ by Lemma 3. If one rectangle is axis-aligned, it is a rival of P, which contradicts (ii). If both are tilted, choose a column c that meets P. Both rectangles have a corner in column c and span at most 3, so they lie in the columns $c-3$ to $c+3$. This contradicts (iii). Hence the candidate is valid. It has $m+4$ points, so it is an optimum. □

By Lemma 9, whether a set P of 1 or 2 white points has a rival depends only on F and on the letters of the columns that meet P. So the 3 conditions can be tested while the word is read from left to right.

Proposition 26 (the automaton)**.** *Let F be the edge set of a 4-cycle on the rows. The automaton $\mathcal{A}_F$ reads words in the 4 singleton letters and the 4 pairs of F. A state (u, U) records the last 6 letters u (all letters, if fewer than 6 have been read) and the set $U \subseteq F$ of pairs read so far. The start state is the empty word with $U = \emptyset$. On reading a letter y, the automaton rejects if $y \in U$, or if some tilted rectangle in the word uy that has a corner in the new column*

(1) is hidden, or

(2) shows 1 or 2 corners and has a rival, or

(3) has the same presentation as another tilted rectangle in uy.

Otherwise it moves to the state given by uy and $U \cup (\{y\} \cap F)$. A word is accepted if it is read to the end without rejection and the final state has $U = F$. For each $m \geq 6$, the accepted words of length m are exactly the optima of the $4 \times m$ grid with pair set F.

Proof. An accepted word contains each pair of F exactly once, so it is a candidate with pair set F. A tilted rectangle has span at most 3. When its last column is read, it lies in the last 4 letters of uy, and it is tested for (1) and (2). Two tilted rectangles in 7 consecutive columns both lie in uy when the later of their last columns is read, and they are compared in (3). So an accepted word of length $m \geq 6$ satisfies the conditions of Lemma 25 and is an optimum. Conversely, an optimum with pair set F is a candidate, so no pair is read twice and the final state has $U = F$. It satisfies the conditions of Lemma 25, so no step rejects. □

5.3 The generating function and the growth rate

The statements of this subsection are computer-assisted. For the 3 pair sets the script `four_row_certificate.py` builds $\mathcal{A}_F$ from Proposition 26. The automata have 60975, 138095, and 55575 states. Write $a_F(m)$ for the number of words of length m that $\mathcal{A}_F$ accepts, so that $a_4(m) = \sum_F a_F(m)$ for $m \geq 6$. At $m = 12$ the pair sets of $\{1,2\} \mid \{3,4\}$, $\{1,3\} \mid \{2,4\}$, and $\{1,4\} \mid \{2,3\}$ give 114424, 1785768, and 137656 optima. For $m \leq 10$ the counts agree with a test of every candidate against the definition.

Theorem 27 (four-row generating function, computer-assisted)**.** *There are coprime integer polynomials $P_4 = 26z^6 - 826z^7 + \cdots$ of degree 444 and Q_4 of degree 420 with $Q_4(0) = 1$ and*

$$\sum_{m \geq 6} a_4(m) z^m = \frac{P_4(z)}{Q_4(z)}.$$

Write $Q_4 = \sum_j q_j z^j$. The recurrence $\sum_{j=0}^{420} q_j \, a_4(n-j) = 0$ holds for every $n \geq 445$ and fails at $n = 444$. Over the integers,

$$Q_4 = (1-z)^2 (1+z)^2 \, g_5^3 \, \tilde{g}_5^3 \, g_6^3 \, g_{11}^5 \, g_{17}^3 \, g_{19}^3 \, g_{20}^5 \, g_{21}^5,$$

where g_d and $\tilde{g}_d$ are irreducible of degree d with constant term 1. Here $g_5 = 1 + 3z^2 - 2z^3 + 2z^4 - z^5$, $\tilde{g}_5 = 1 + z^2 - z^5$, $g_6 = 1 - z^2 - z^5 - 2z^6$, and

$$\begin{aligned} g_{20} = {} & 1 - 4z + 2z^2 + z^3 - 3z^4 + 5z^5 + z^6 - 3z^7 + 7z^8 + 2z^9 + 30z^{10} - 7z^{11} + 10z^{12} \\ & + 12z^{13} + 2z^{14} + 17z^{15} + 26z^{16} + 19z^{17} - 26z^{18} - 4z^{19} + 8z^{20}. \end{aligned}$$

Proof. The polynomials P_4 and Q_4 were first found from the values $a_4(m)$ with $m \leq 1500$ by the Berlekamp–Massey algorithm [8]. This is only a guess, so we certify it on the automata $\mathcal{A}_F$. For a state s of $\mathcal{A}_F$ let $u_F(n)_s$ be the number of words of length n that lead from s to acceptance. Then $a_F(n)$ is the entry of $u_F(n)$ at the start state, and $u_F(n+1) = T_F u_F(n)$ for the transition matrix T_F. The script checks that the vector $R_F = \sum_j q_j \, u_F(445-j)$ is zero for each F. Applying T_F^{n-445} gives $\sum_j q_j \, a_F(n-j) = 0$ for all $n \geq 445$, and we sum over F. The number $a_F(n)$ counts optima only for $n \geq 6$, and this recurrence uses only the indices $n - j \geq 25$. The check is exact for the following reason. The script works modulo each of 48 primes below 2^{31}. An entry of $u_F(n)$ is at most 8^n, so an entry of R_F has absolute value at most $8^{445} \sum_j |q_j|$. This is less than half the product of the primes. So R_F vanishes because it vanishes modulo every prime. The script also checks, modulo the same primes, that the product of Q_4 and $\sum_{m=6}^{444} a_4(m) z^m$ agrees with P_4 up to z^{444}. Here $a_4(m) \leq 3 \cdot 8^m$, so a coefficient of the product differs from the coefficient of P_4 by at most $3 \cdot 8^{444} \sum_j |q_j|$ plus the largest coefficient of P_4 in absolute value. This is again less than half the product of the primes, so the agreement is exact. The coefficient of z^{444} in P_4 is not zero, so the recurrence fails at 444. The factorization and the coprimality were computed with SymPy [10]. □

All coefficients of P_4 and Q_4 and all factors of Q_4 are in the ancillary file `four_row_gf.txt`, which is also attached to the OEIS entry A401224. The 3 series $\sum_m a_F(m) z^m$ are rational, with reduced denominators of degrees 162, 153, and 107. The factor g_{20} occurs only for the pair set of $\{1,3\} \mid \{2,4\}$.

Corollary 28 (growth, computer-assisted)**.** *Let* $\rho_4 = 3.351996777310\ldots$ *be the largest real root of* $x^{20}g_{20}(1/x)$*. There is a polynomial* π *of degree* 4 *with* $a_4(m) = \rho_4^m\,\pi(m) + O\big((1000/341)^m\big)$*. Hence* $a_4(m) \sim c\,m^4\rho_4^m$ *with* $c = 8.16109707\ldots \times 10^{-5}$*, and almost all optima have the pair set of* $\{1,3\} \mid \{2,4\}$*.*

Proof. We isolate the complex roots of every factor of Q_4 with exact rational bounds. The only root of modulus at most 0.341 is $r = 1/\rho_4 = 0.29832\ldots$. It is a simple root of g_{20}, so it is a pole of order 5 of P_4/Q_4, because P_4 and Q_4 are coprime. Write $P_4/Q_4 = \sum_{j=1}^{5} c_j(1 - z/r)^{-j} + T(z)$ with $c_5 \neq 0$. The series T has no pole of modulus at most 0.341, so its coefficients are $O((1000/341)^m)$. Since $[z^m](1 - z/r)^{-j} = \rho_4^m\binom{m+j-1}{j-1}$, we get $\pi(m) = \sum_j c_j\binom{m+j-1}{j-1}$ and $c = c_5/24$. Interval arithmetic gives $8.1610970775 \times 10^{-5} < c < 8.1610970776 \times 10^{-5}$. The last claim holds because g_{20} does not divide the denominators of the other 2 pair sets. The order 5 of the pole matches the 5 runs of singletons that the 4 doubles separate. □

6 Five rows

Let $m \geq 8$. By Theorem 10, $f(5,m) = m + 6$, and an optimum has 6 doubles, whose row pairs are the edges of a complete bipartite graph with parts of sizes 2 and 3, and $m - 6$ singletons. Table 3 lists the number of optima for $8 \leq m \leq 23$ (A401225). These values are computer-assisted. Two separately written transfer counters over the columns agree for $13 \leq m \leq 23$, and for $8 \leq m \leq 12$ the values agree with complete enumerations (Section 8).

m	$a_5(m)$	m	$a_5(m)$	m	$a_5(m)$	m	$a_5(m)$
8	4	12	24706	16	52128242	20	45743165894
9	44	13	193406	17	299349210	21	230934909328
10	350	14	1346758	18	1653637492	22	1139486126866
11	3004	15	8650416	19	8828301226	23	5510071794736

Table 3: The number of optima of the $5 \times m$ grid (computer-assisted).

7 Every fixed height

Fix the height h. The letters are now all subsets of $\{1,\ldots,h\}$, and a blackout of the $h \times m$ grid is the column word $w = B_1 \cdots B_m$. Write $|w| = m$ for its length and $S(w)$ for the blackout. Let $\mathcal{L}_h$ be the set of words of valid blackouts, of all lengths. This section does not use the companion paper.

Theorem 29 (regularity)**.** *For every* $h \geq 1$ *the language* $\mathcal{L}_h$ *is regular. Hence the series* $\sum_{w\in\mathcal{L}_h} z^{|w|}u^{|S(w)|}$ *is a rational function of* z *and* u*.*

Proof. The *trace* of a rectangle of the $h \times m$ grid is the word $Y_1 \cdots Y_m$, where Y_i is the set of rows of its corners in column i. Different rectangles have different traces. By Lemma 2,

the word w is not valid if and only if there are 2 different traces Y and Y' of length m with $Y_i \triangle Y'_i \subseteq B_i$ for every i.

The traces of all lengths form a regular language. The trace of an axis-aligned rectangle is a letter $\{r, s\}$ with $r \neq s$, then some empty letters, then the same letter again, with any number of empty letters before and after. By Lemmas 4 and 5, the trace of a tilted rectangle is one of finitely many words of length at most $W_h + 1$, again with empty letters before and after.

Read a triple (w, Y, Y') of words of the same length as one word whose letters are triples of sets. The triples in which Y and Y' are different traces and $Y_i \triangle Y'_i \subseteq B_i$ for every i form a regular language. Forgetting Y and Y' keeps it regular. So the words that are not valid form a regular language, and $\mathcal{L}_h$ is its complement.

Take a deterministic automaton for $\mathcal{L}_h$ and give the step that reads B the weight $zu^{|B|}$. Let $T(z, u)$ be the matrix of these weights, α the indicator row vector of the start state, and β the indicator column vector of the accepting states. Each word of $\mathcal{L}_h$ has exactly 1 accepting path, so the series is $\alpha(I - T(z, u))^{-1}\beta$. This is rational by Cramer's rule [3, Proposition V.6]. □

The *excess* of a word is $e(w) = |S(w)| - |w|$, so that $e(uv) = e(u) + e(v)$ and $f(h, m) - m$ is the largest excess of a word of length m in $\mathcal{L}_h$.

Theorem 30 (fixed height)**.** *Let $h \geq 1$.*

(a) For every integer k the words of $\mathcal{L}_h$ with excess k form a regular language.

(b) The sequence $f(h, m) - m$ is ultimately periodic in m.

(c) The column words of the optima of the grids with h rows form a regular language, and $\sum_{m \geq 1} a_h(m) z^m$ is a rational function.

Proof. (a) We first bound the excess of a valid word v. If $|v| \leq 2$, then $e(v) \leq 2h - 2$. Let $|v| \geq 3$. By Lemma 8 no 2 columns share 2 black rows, so $\sum_i \binom{|B_i|}{2} \leq \binom{h}{2}$. Since $|B_i| - 1 \leq \binom{|B_i|}{2}$, we get $e(v) \leq \binom{h}{2}$. Put $c_h = \max\left(\binom{h}{2}, 2h - 2\right)$. Every factor of a valid word is the word of a valid blackout of a smaller grid. Now let $w = uv$ be valid with $e(w) = k$. Then

$$k - c_h \leq e(u) = k - e(v) \leq c_h.$$

So an automaton for $\mathcal{L}_h$ can also keep the excess of the prefix read so far. It rejects when this value leaves the interval $[k - c_h, c_h]$, and at the end it accepts when the value is k. This is a finite automaton for the words of $\mathcal{L}_h$ with excess k.

(b) Let Λ_k be the set of lengths of the words in (a). Replacing every letter by the same symbol turns a regular language into a regular language over 1 letter, and a deterministic automaton over 1 letter is a path that ends in a cycle. So each Λ_k is ultimately periodic. The blackout that consists of row 1 is valid. Indeed a tilted rectangle has only 1 corner in its top row, as in Lemma 4. So a rectangle with fewer than 3 white corners is axis-aligned with a side in row 1, and its 2 white corners determine it. Hence $f(h, m) - m$ is the largest k with $0 \leq k \leq c_h$ and $m \in \Lambda_k$. These are finitely many ultimately periodic sets, so $f(h, m) - m$ is ultimately periodic.

(c) For each k let D_k be the set of all $m \geq 1$ with $f(h, m) - m = k$. It is ultimately periodic by (b), so the words with length in D_k form a regular language. The words of the optima are the words of $\mathcal{L}_h$ with excess k and length in D_k, for $0 \leq k \leq c_h$. This is a finite union of regular languages. Its series is rational, as in the proof of Theorem 29. □

By the companion paper [12], the sequence $f(h, m) - m$ is in fact eventually constant, equal to q_h for $h \geq 3$. For the largest excess that occurs, part (a) also follows from a result of Guilhot, Little, and Parkinson [4, Theorem 1(3)] on additive weights that are bounded above on a regular language.

8 Computations and verification

The code and the stored results are in the repository [13]. From its root, the scripts are in `jis/scripts` and the evidence files in `jis/data`, and `jis/scripts/README.md` gives the commands. The ancillary files of this paper are in `papers/enumeration/anc`. Every computed count has 2 separately written checks.

Three rows. The proofs in Section 4 are by hand up to the expansion of rational functions, which we did with SymPy [10]. One program finds all optima of the $3 \times m$ grid from Definition 1 for $m \leq 10$, and its counts agree with (11). A transfer matrix built from Table 2 agrees with (11) and (13) for $m \leq 100$. For Theorem 22, the fixed words are listed for $m \leq 24$ and the orbits for $m \leq 10$. The files are `three_row_*.json`.

Four rows. The ancillary script `four_row_certificate.py` builds the automata of Proposition 26 and carries out the proof of Theorem 27 in about 5 minutes. It needs only `four_row_gf.txt`. The repository also holds an earlier computation that does not use this script (`four_row_*.json`). It works with smaller deterministic machines found by a search, checks on the product automaton that they accept the same words as $\mathcal{A}_F$, and gives the same counts for all $m \leq 1500$ with 2 transfer counters. It also provides the series of the 3 pair sets and the isolation of the roots of Q_4.

Five rows and small grids. The files for Table 3 are `five_row_count_*.json` and `five_row_optima_dp_w*.json`. The counts $a_4(4)$, $a_4(5)$, $a_5(5)$, $a_5(6)$, and $a_5(7)$ of Table 1 come from complete enumerations, each done by 2 programs (`verify_narrow.py` and `table_counts`). The SAT enumerations use PySAT [5]. Each blackout found is tested again against Definition 1.

Formal proofs. The folder `lean` of the repository contains proofs in Lean of Lemmas 2, 3, 5, and 8, of the Strip Theorem with its count, and of the example after Theorem 23.

9 Open questions

(1) What is the rational series $\sum_m a_5(m)z^m$, and what is the growth rate of $a_5(m)$?
(2) Is $a_h(m) \sim c_h\, m^{q_h} \rho_h^m$ for every h, as it is for $2 \le h \le 4$?
(3) Remark 6 settles the largest entry in each row of A338885. Is its smallest entry in row n equal to A228286(n), as conjectured there?

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